

Notes: Dasaratha (RES, 2023)

Innovation and Strategic Network Formation

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1 Big picture

- **Question.** How do firms choose openness versus secrecy when innovation requires combining ideas and ideas can spread through a network?
- **Core trade-off.** Openness lets a firm learn ideas from others, but it also lets other firms learn its own ideas and compete.
- **Main theorem.** In large economies, every investment equilibrium is critical: the learning network is at the threshold between sparse and dense networks.
- **Policy message.** Increasing openness can have much larger effects than increasing R&D alone.

2 Incremental contribution of the paper

- Develops a tractable theory of endogenous random network formation with continuous openness choices.
- Connects strategic openness and secrecy to random-graph phase transitions.
- Microfounds knowledge spillovers through combinatorial innovation and indirect learning.
- Shows that equilibrium innovation can vanish even though slightly more openness produces non-vanishing innovation.
- Provides a formal role for public innovators and a nuanced view of patent rights as information intermediaries.

3 Model objects and measurement

There are n firms. Firm i can discover idea i with probability p_i and pays cost $c(p_i)$. A technology is a set of k ideas. Firm i chooses openness q_i , and the baseline interaction rate is

$$\iota(q_i, q_j) = q_i q_j.$$

A firm earns monopoly profit from a technology only if it knows all component ideas and no competitor knows all of them.

4 Models and theoretical specifications

4.1 Payoffs

$$U_i(p, q) = E[|PT_i(p, q)|] - c(p_i).$$

4.2 Criticality

In symmetric equilibrium,

$$\delta \iota(q^*, q^*) n \approx 1.$$

Theorem 1 states that for large n , symmetric equilibria exist and every investment equilibrium is critical.

4.3 Discovery rate

$$D(p, q) = \frac{E[|\cup_i T_i(p, q)|]}{\binom{n}{k}}.$$

At equilibrium $D(p^*, q^*) \rightarrow 0$, but for any $\epsilon > 0$, $D(p^*, (1 + \epsilon)q^*)$ is non-vanishing.

5 Identification and theoretical design

This is a theory paper. The mechanism is identified mathematically by isolating how endogenous openness choices interact with random network phase transitions. The criticality result relies on indirect learning, combinatorial technologies, and the endogenous competition cost of outgoing links.

6 Section-by-section study guide

- **Introduction.** Explains why openness can raise innovation and why firms may choose secrecy.
- **Model.** Defines ideas, technologies, private investment, openness, indirect learning, and proprietary technologies.
- **Equilibrium.** Establishes that all equilibria are critical.
- **Welfare.** Shows large gains from increasing openness and limited gains from R&D subsidies alone.
- **Extensions.** Studies public innovators, heterogeneity, payoff structures, and patents.

7 Tables and figures

7.1 Figure 1: Network with four firms and $k = 3$

Read: Figure 1 shows a network in which Firm 3 uniquely knows all ideas in technology $t = \{1, 3, 4\}$ and earns monopoly profit.

Detailed interpretation. The figure demonstrates how indirect learning creates proprietary technologies.

Economic message. Indirect learning is valuable when it gives one firm a unique complete combination of ideas.

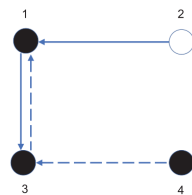


FIGURE 1
Network with four firms and $k = 3$. Black circles are firms that discover ideas while white circles do not discover ideas. Arrows are in the direction of information flow, and dashed arrows indicate direct learning while solid arrows indicate indirect learning. The only technology produced is $t = \{1, 3, 4\}$ and Firm 3 receives monopoly profit.

Network with four firms and $k = 3$

7.2 Figure 2: Network with additional indirect learning

Read: Firm 1 also learns through Firm 3, so both Firm 1 and Firm 3 can produce $t = \{1, 3, 4\}$.

Detailed interpretation. The same diffusion that creates technologies can also eliminate monopoly profits.

Economic message. Openness has a private competition cost and a social learning benefit.

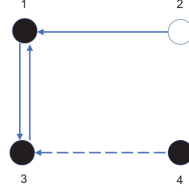


FIGURE 2

Network with four firms and $k = 3$. Black circles are firms that discover ideas while white circles do not discover ideas. Arrows are in the direction of information flow, and dashed arrows indicate direct learning while solid arrows indicate indirect learning. Firm 1 now learns indirectly from Firm 3, unlike in Figure 1. The only technology produced is $t = \{1, 3, 4\}$, and there are no profits because Firms 1 and 3 both produce t .

Network with additional indirect learning

7.3 Table 1: Best responses and policy implications

Read: The table summarizes subcritical, critical, and supercritical regions. Discovery is vanishing at equilibrium but non-vanishing in the supercritical region.

Detailed interpretation. R&D subsidies have small effects at equilibrium but large effects once openness is above equilibrium.

Economic message. Policy should first raise interaction, then subsidize R&D.

TABLE 1
Best responses and policy implications when firms choose symmetric strategies (p, q) in the subcritical, critical, and supercritical regions

	Subcritical	Critical	Supercritical
Best response q_i	High	Intermediate	Low
Discovery rate	Vanishing	Vanishing	Non-vanishing
Increasing q	Large benefit	Large benefit	Ambiguous
Increasing p	Small benefit	Intermediate	Large benefit

Best responses and policy implications

8 Short summary

- Firms create technologies by combining ideas.
- Openness creates learning benefits and competition costs.
- The equilibrium network is critical.
- Equilibrium discovery vanishes asymptotically.
- Slightly more openness creates a non-vanishing discovery rate.
- Public innovators can serve as information intermediaries.

9 Expanded conclusion

9.1 Core conclusion

Dasaratha shows that strategic secrecy places innovation networks at the edge of large-scale diffusion.

9.2 Economic intuition

In sparse networks, firms want more openness to access structural holes. In dense networks, firms want less openness because competition erases monopoly profits. Equilibrium lies at the boundary.

9.3 Incremental contribution

The paper links endogenous random network formation, phase transitions, and combinatorial innovation in one tractable framework.

9.4 Policy implication

Policies that raise interaction, such as innovation clusters, open science, public innovators, or weaker non-competes, can be more powerful than R&D subsidies alone.

9.5 Limitations

The model is theoretical and asymptotic and abstracts from many contracting and institutional details.

9.6 Takeaway

Innovation policy should target the architecture of learning networks, not only the amount of private R&D.